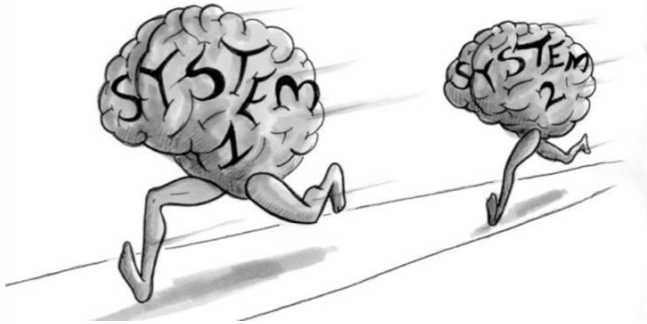
# The Neuroscience of Reward & Learning

From Canonical Models to Contemporary Circuit Dynamics

# Part 1: The Foundations of Value

How we make choices, represent value, and compute basic associations.

# Dual Process Theory




## System 1 (Model-Free)

Fast

Automatic

Computationally efficient

Inflexible

Unreasonable

## System 2 (Model-Based)

Slow

Deliberate

Flexible

Effortful

Logical

# Behavioral Control

## The Habitual Circuit

Fast

Automatic

Indifferent to outcome

Inflexible

Unreasonable

## The Goal-Directed Circuit

Planned course of action

Deliberate

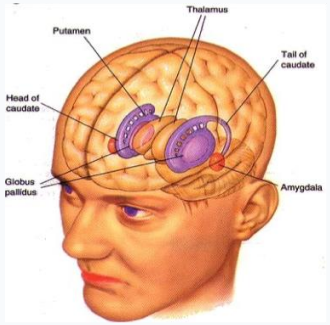
Sensitive to outcome value

Flexible

Logical

# Behavioral Control

## Model Free control

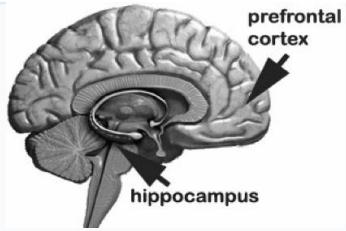



Automatic

Slow to learn

Cheap to use

implemented in Basal ganglia

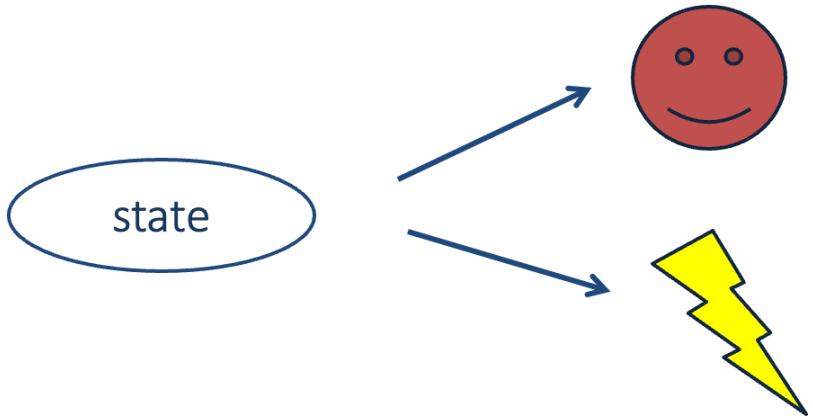



## Model-Based control

Model-based, forward-planning computations rely heavily on value representations within the prefrontal cortex, hippocampus, and the dorsomedial striatum.

# The Machine Learning Landscape

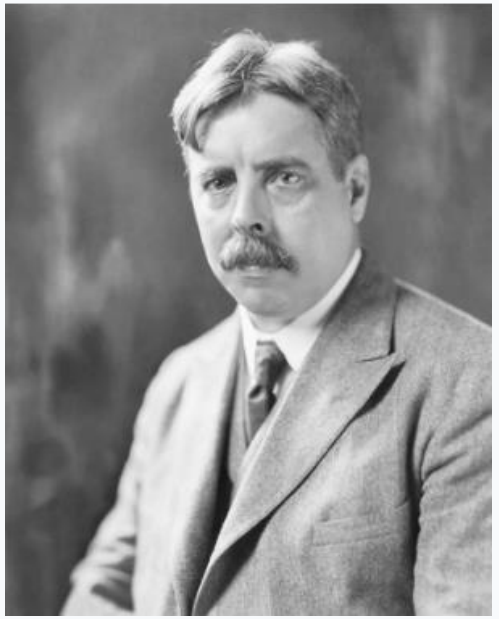
- **Supervised Learning:** Relies on an all-knowing teacher providing detailed, explicit feedback.
- **Unsupervised Learning:** Driven by self-organization and inherent data structure without explicit labels.
- **Reinforcement Learning:** Driven by scalar (correct/incorrect) feedback to evaluate and optimize state-action pairings.



# Historical Foundations

## The Law of Effect

**E.L. Thorndike (1911):** Responses accompanied by satisfaction become firmly connected to the situation, increasing their likelihood of recurrence.



*E.L. Thorndike*

> "The Law of Effect is that: Of several responses made to the same situation, those which are accompanied or closely followed by satisfaction to the animal will, other things being equal, be more firmly connected with the situation, so that, when it recurs, they will be more likely to recur"

Edward Lee Thorndike (1911)

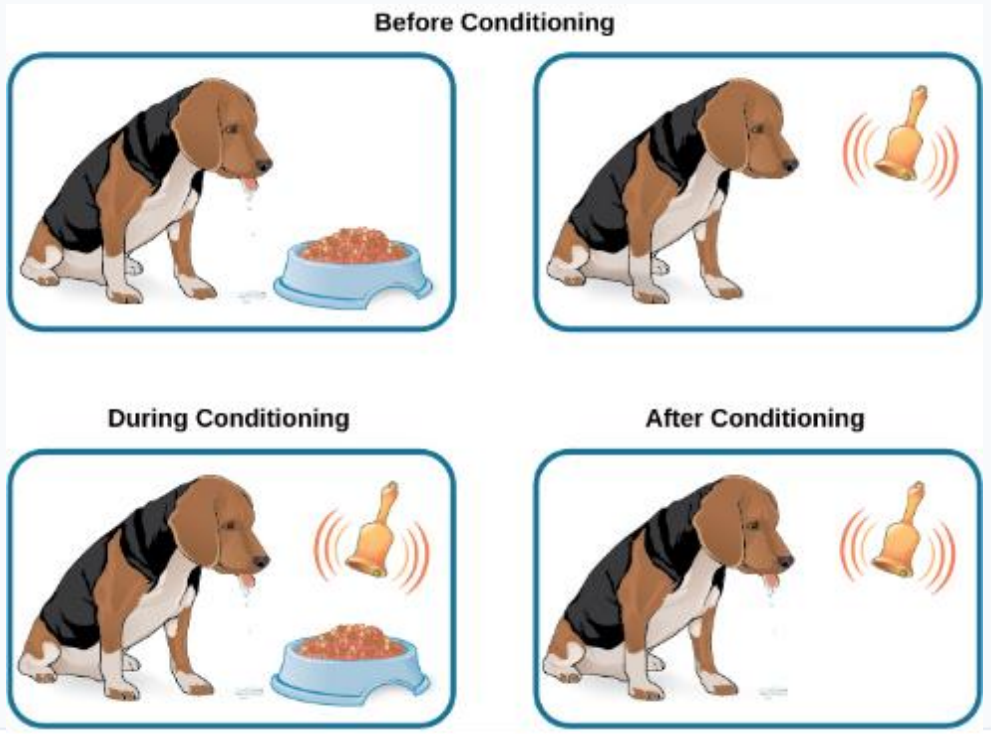
# Early attempts at modeling

STIMULUS — RESPONSE — ± OUTCOME

## By Associative rules

## Classical conditioning

**Ivan Pavlov (1927):** Learning through associative rules where a Conditioned Stimulus (CS) becomes linked to an Unconditioned Stimulus (US) to produce a Conditioned Response (CR).



# Pavlovian conditioning properties

- Acquisition.
- Partial Reinforcement (probabilistic).
- Generalization.
- Inter-Stimulus-Interval (ISI) effects.
- Inter-trial-Interval (ITI) effects

A simple association (coincidence, Hebbian)

model can explain the phenomenon

# Pavlovian conditioning – more properties

- Conditioned  Inhibition (Rescorla 1969)
  - (1) CS1+US→CR  (2) CS1+CS2+ noUS (3) CS2+US→ ~~CR~~
- Blocking (Kamin 1968)
  - (1) CS1+US→CR  (2) CS1+CS2+ US (3) CS2+US→ ~~CR~~
- Latent Inhibition
  - (1) CS1 +noUS (2..k) CS1 +noUS (k) CS1+US → ~~CR~~
- Relative validity (Wagner 1968).
- ...

A simple association (coincidence, Hebbian)

model cannot explain all phenomena

# The Limits of Hebbian Co-occurrence

## Simple Association

A simple associative (Hebbian) model successfully explains the basic acquisition of behaviors, stimulus generalization, and partial reinforcement effects.

## The Problem

Simple association fails to account for complex behavioral phenomena:

- ✕ Conditioned Inhibition (Rescorla, 1969)
- ✕ Blocking (Kamin, 1968)
- ✕ Latent Inhibition

# The Necessity of Surprise

> Learning does not occur merely because two events co-occur. Learning occurs because the co-occurrence is otherwise unpredicted.

# The Necessity of Surprise

"Learning occurs when the world violates expectations CS must RELIABLY predict US"

# The Rescorla-Wagner Model (1972)

$$RPE = R - V$$

$$V(s) \leftarrow V(s) + \eta RPE$$

## Formalizing Expectations

- **Identify the State:** $s = X$
- **Form Prediction:** $V(s) = wX$
- **RPE > 0:** Increase $V$ (Better than expected)
- **RPE < 0:** Decrease $V$ (Worse than expected)
- **RPE = 0:** State remains unchanged

After learning V=R

# Solving the Blocking Effect

## Phase 1: Full Learning

Given stimuli $X_1$ $X_2$ and and reward $R$:

After the value of $X_1$ is fully learned, $V_1 = R$.

## Phase 2: Compound Stimulus

When presented together, the total prediction is $V = w_1 X_1 + w_2 X_2$

Because $RPE = R - V = 0$, no learning occurs for the new weight $w_2$.

# The Rescorla–Wagner Model

## Expectation is Key

Learning is driven not by the mere pairing of a stimulus and a reward, but by the discrepancy between what the animal expects and what it actually receives.

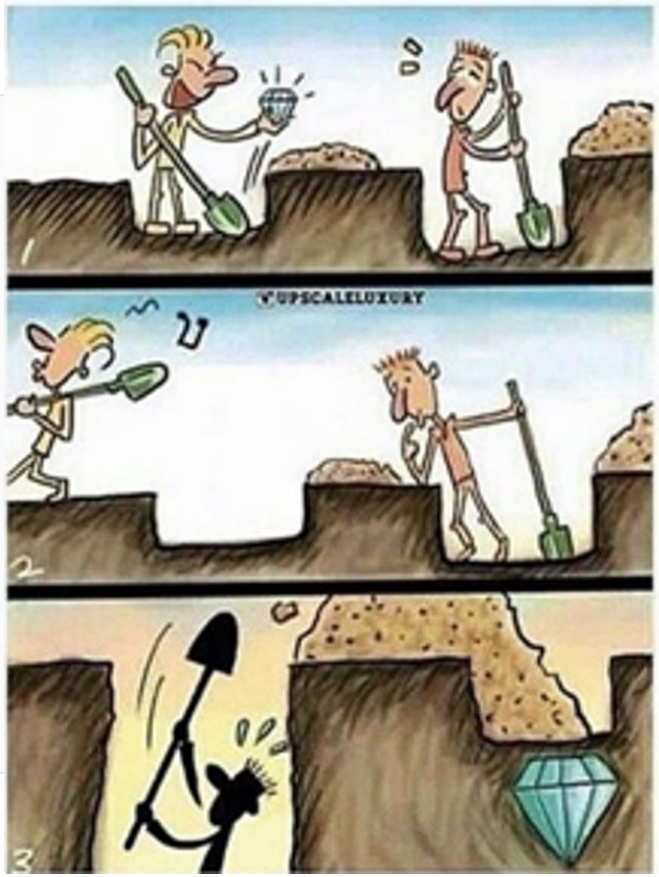
## The Updating Signal

This discrepancy serves as an updating signal. If the reward perfectly matches expectation, no new learning occurs, regardless of how many times the pairing is repeated.

# Implications of Prediction Error

- **Blocking:** If a reward is fully predicted by one cue, adding a second cue will yield no learning for the new cue.
- **Extinction:** Removing the reward generates a negative prediction error, actively unlearning the previous value association.
- **Overshadowing:** More salient cues acquire value faster due to a higher learning rate (alpha).
- **Biological Necessity:** The brain requires a dedicated physical mechanism to compute and broadcast this error globally.

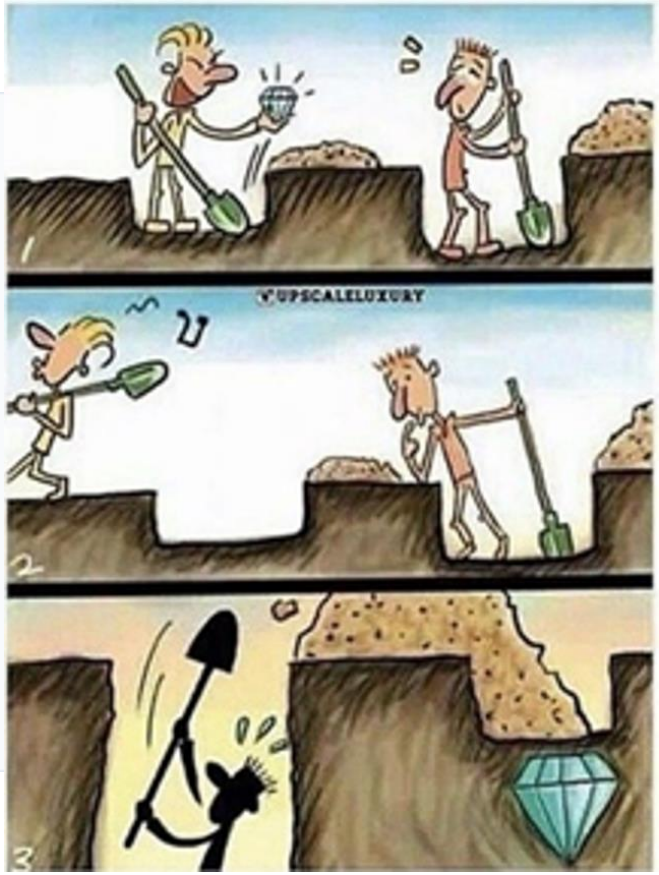
# Adding Actions to the Equation (Q-Learning)




## State-Action Values

- **Prediction:** $Q(s, a) = wX$
- **Action Selection:** Take action $a$ based on probability function $P(a) = f(Q)$
- **RPE > 0:** Increase $Q$, raising action probability
- **RPE < 0:** Decrease $Q$, lowering action probability

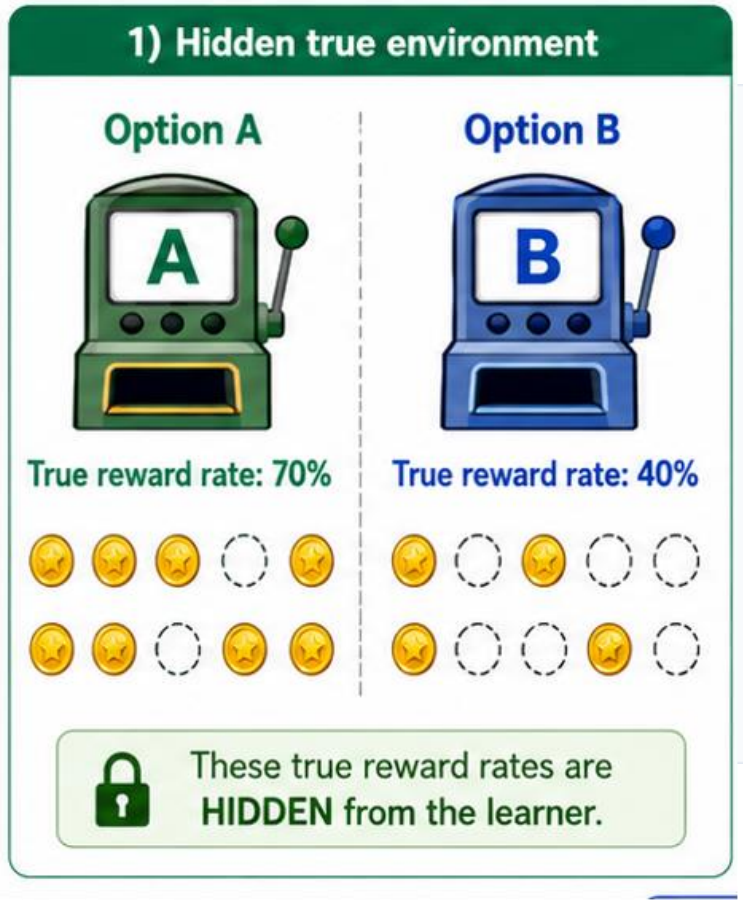
# Exploration-exploitation tradeoff




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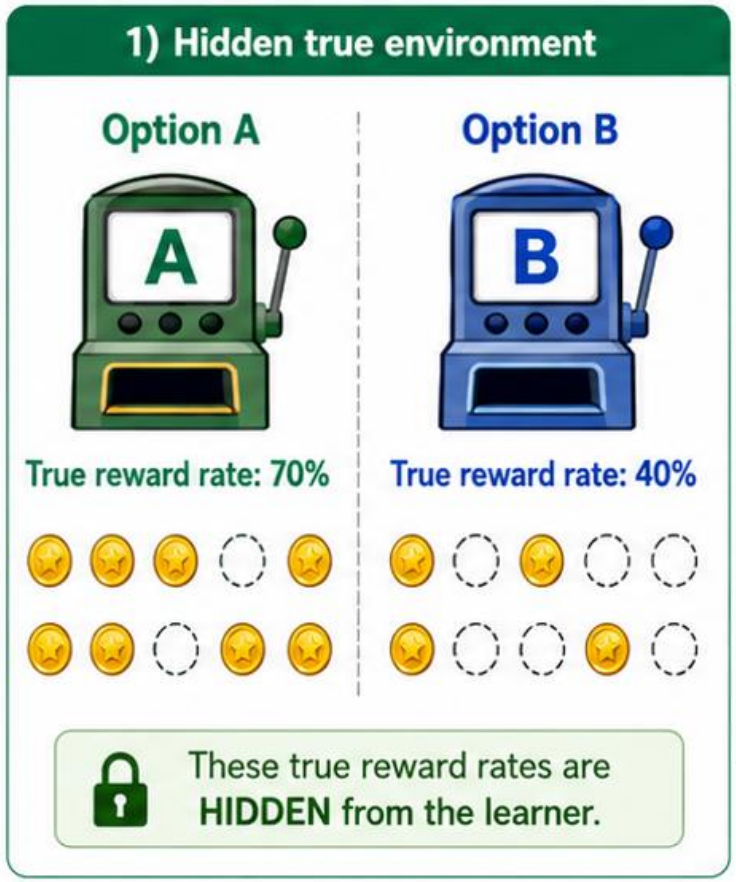
# Exploration-exploitation tradeoff - K armed Bandit

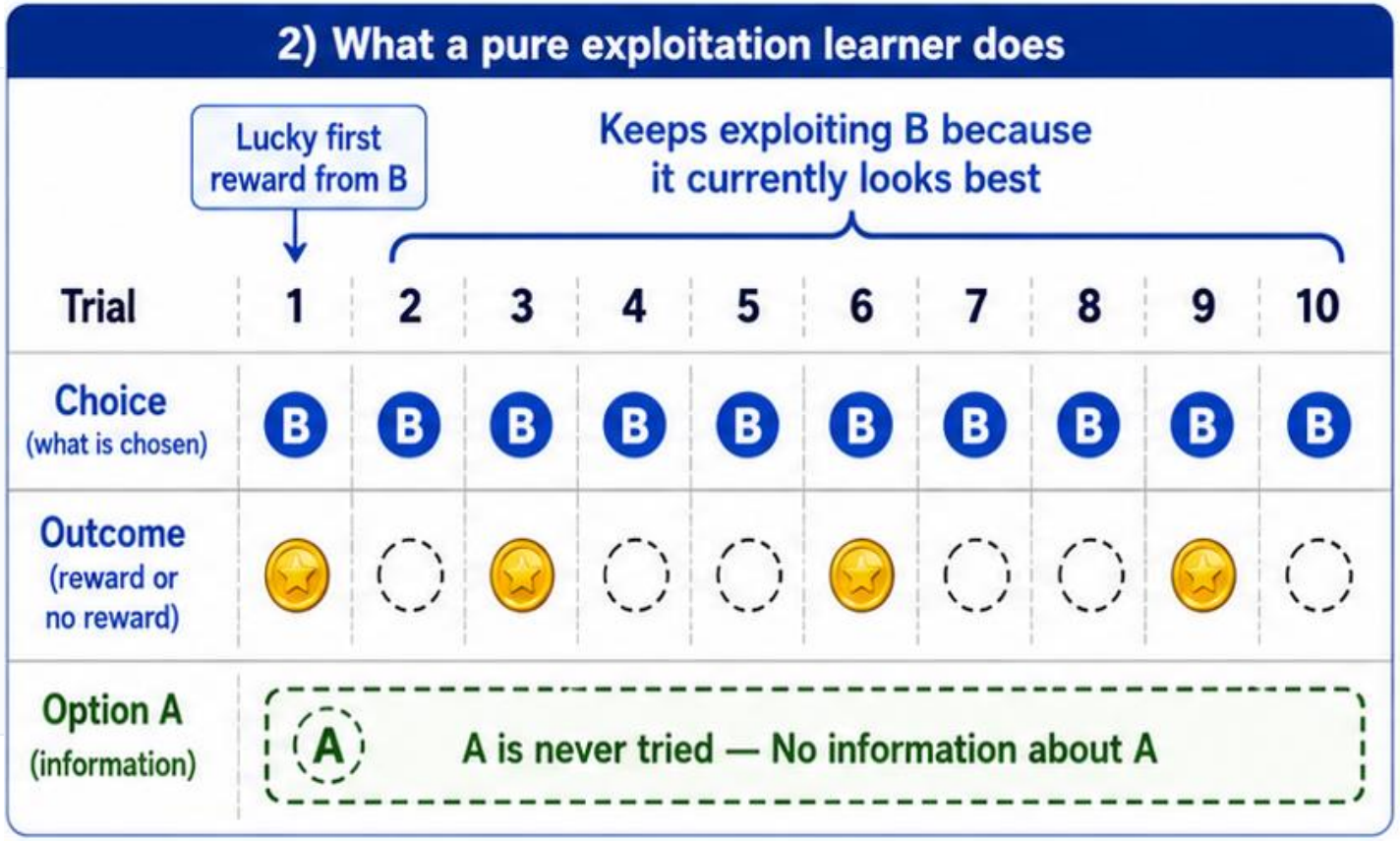



## State-Action Values

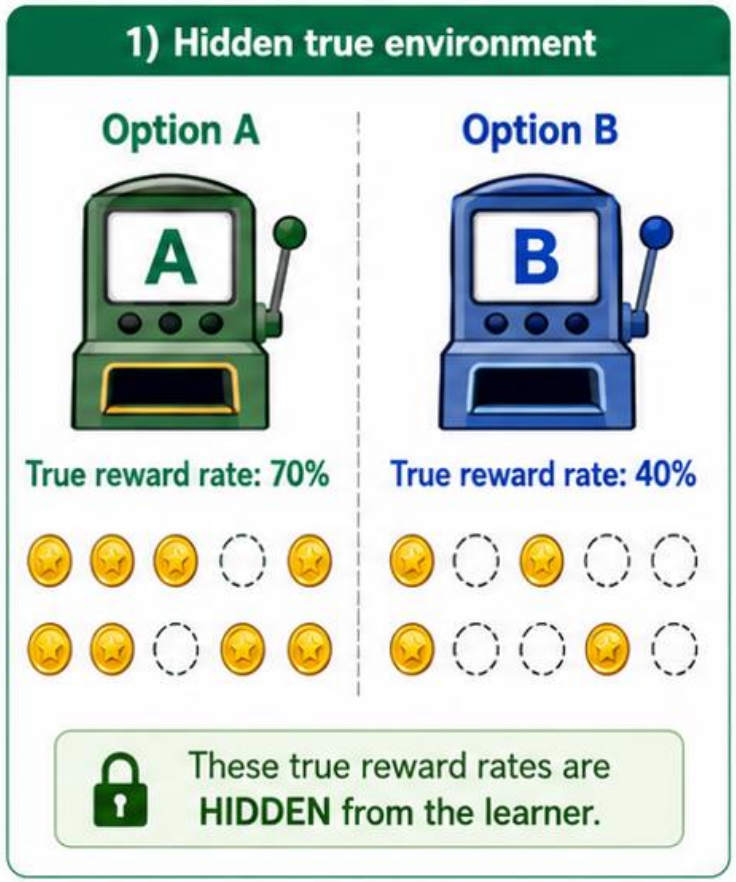
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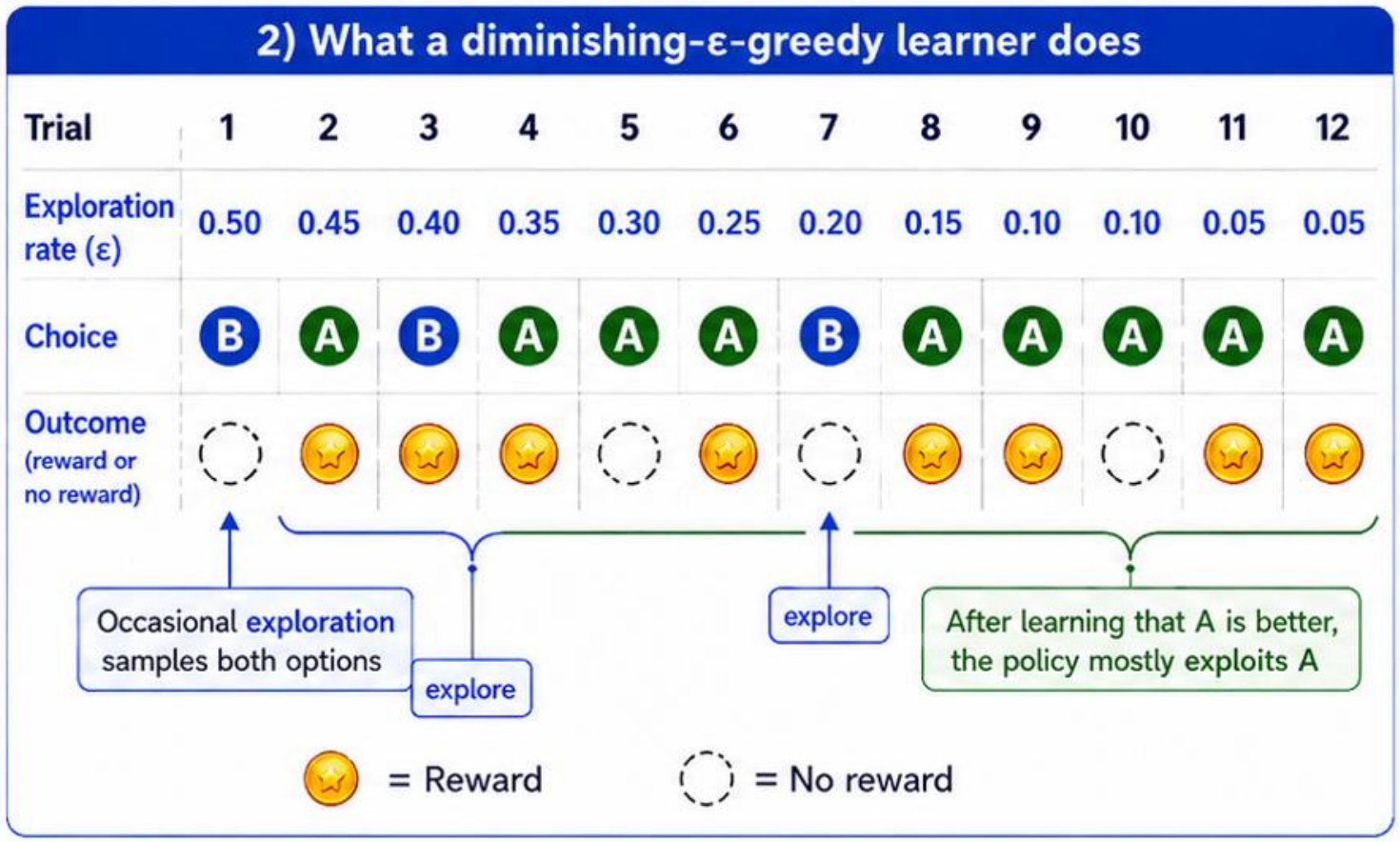
# Exploration-exploitation tradeoff - K armed Bandit



# Solution – introduce randomness



# Softmax policy

$$P(a) = \frac{e^{\beta Q(s,a)}}{\sum_{a'} e^{\beta Q(s,a')}}$$

## State-Action Values

- **Prediction:** $Q(s,a) = wX$
- **Action Selection:** Take action $a$ based on probability function $P(a) = f(Q)$
- **RPE > 0:** Increase $Q$, raising action probability
- **RPE < 0:** Decrease $Q$, lowering action probability

# Prediction error learning

$$RPE = R - V$$

$$V(s) \leftarrow V(s) + \eta RPE$$

## The Need for Prediction Error

The classic Rescorla-Wagner model formally established that associative learning requires a co-occurrence to be unpredicted.

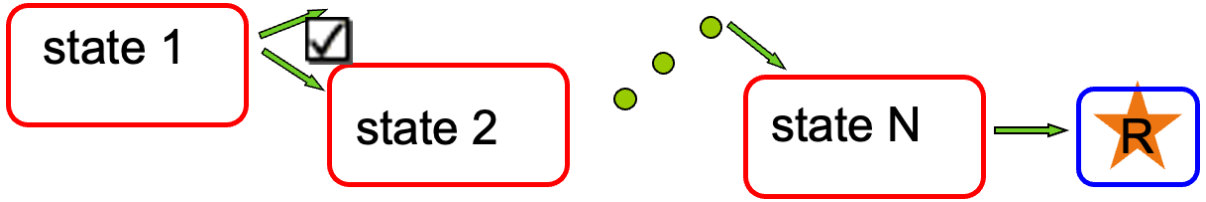
The resulting prediction error acts as the driver for updating the subjective value of any given state.

# Temporal credit assignment problem

$$RPE = R - V$$

$$V(s) \leftarrow V(s) + \eta RPE$$

**Valuation (and actions) are separate from outcome**



# Temporal Difference (TD) learning

*Sutton, 1988*




## Solution to temporal credit assignment

A fundamental challenge in computational RL is correctly associating a present action with a delayed reward.

Sutton and Barto's Temporal Difference (TD) learning solves this dilemma by allowing the system to update value estimates continuously based on successive predictions.

1. Estimate value of current state ($V_t=r_t+ \gamma r_{t+1} + \gamma^2 r_{t+2}$...):

(discounted) sum of expected rewards

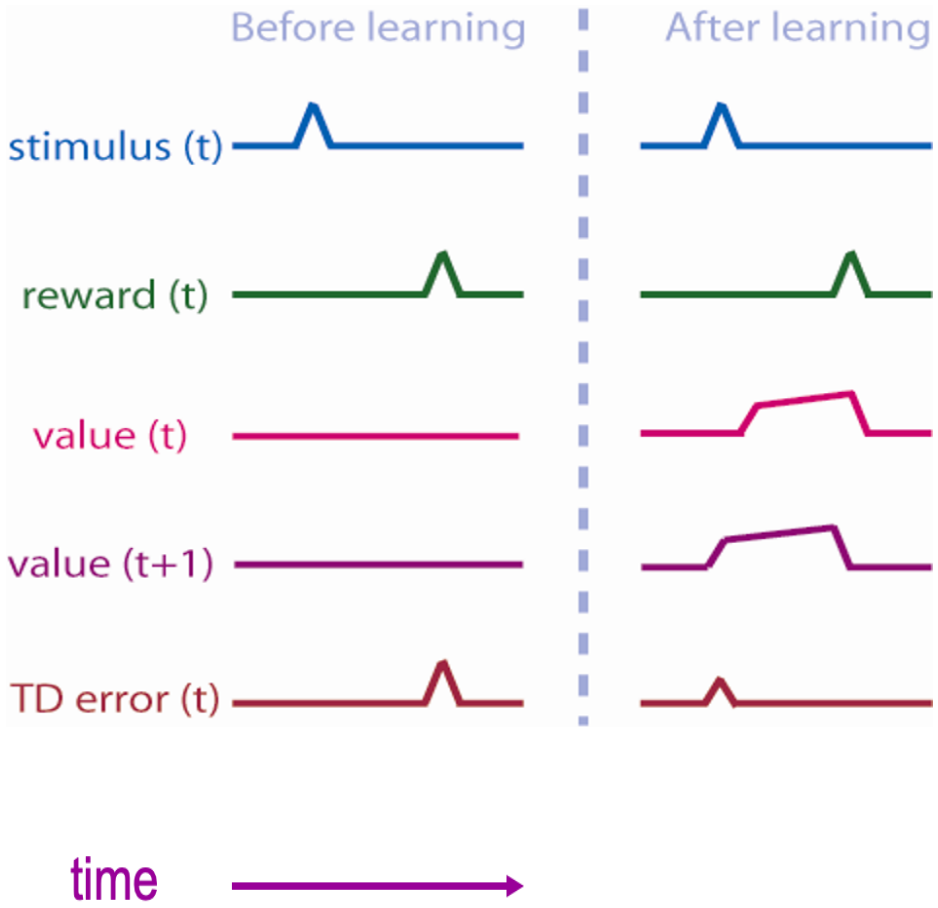
2. Compare current prediction with 'truer' value of current state: reward at present state + estimated value of next state (rt+ γVt+1)

3. Compute TD error $\delta_t = r_t + \gamma V_{t+1} - V_t$

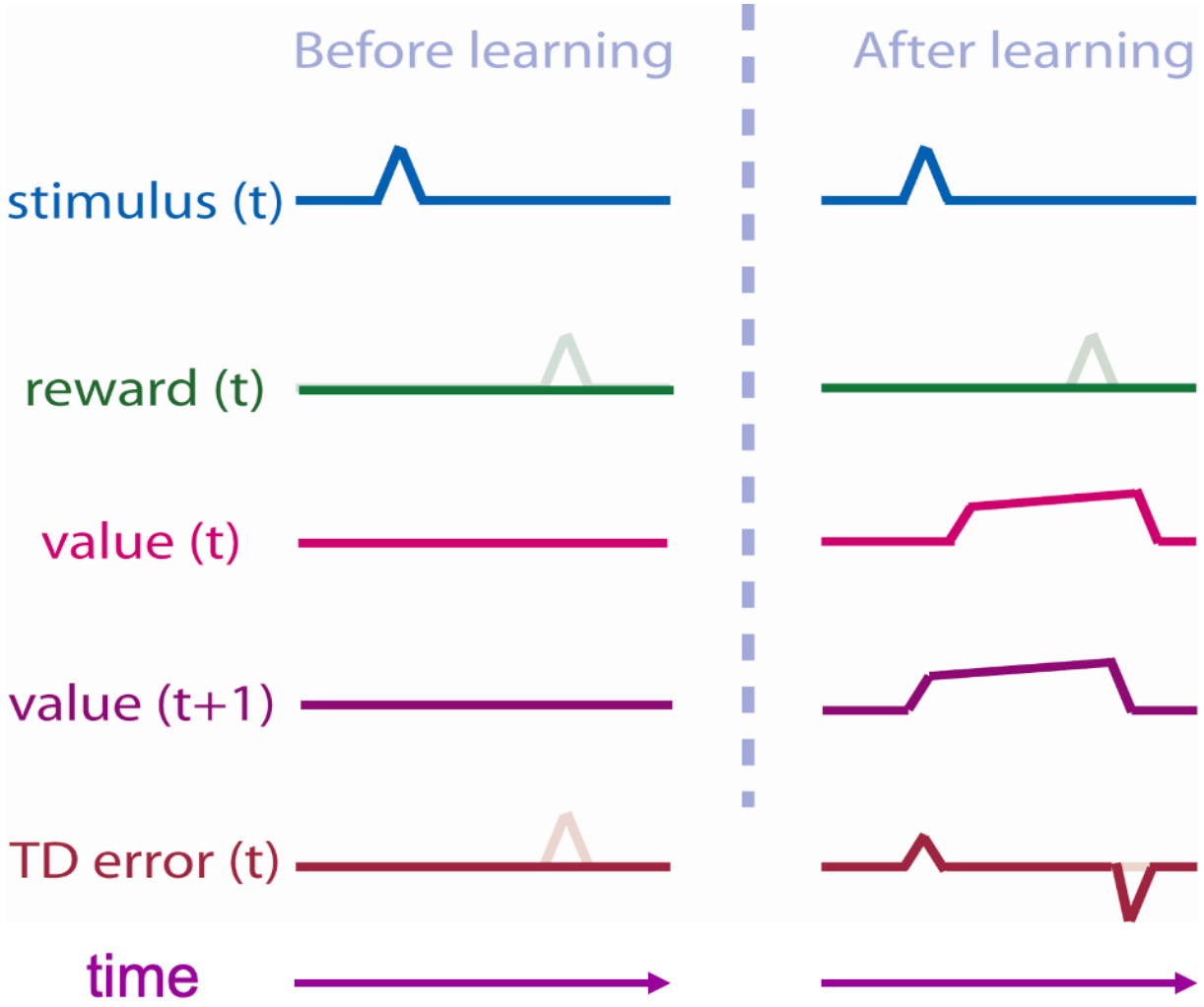
4. Use TD error to improve 1 (Vtk+1=Vtk+η δt)

# TD learning

$$\delta_t = r_t + \gamma V_{t+1} - V_t$$



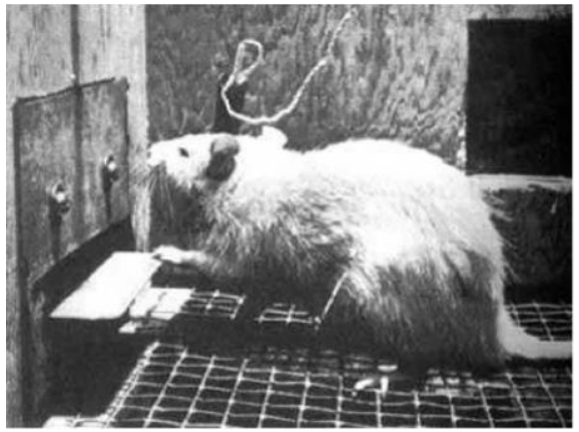
# TD learning


Before learning
After learning
stimulus (t)
reward (t)
value (t)
value (t+1)
TD error (t)
time

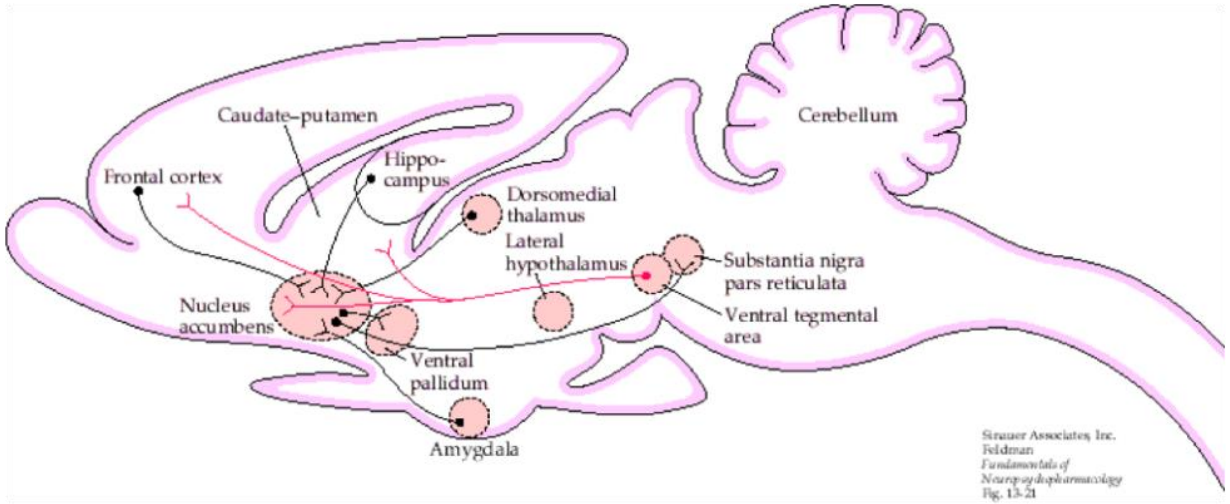
# Part 2: Neurobiology of Canonical RPE

Dopamine, reward and reinforcement learning

# Dopamine, reward and reinforcement learning

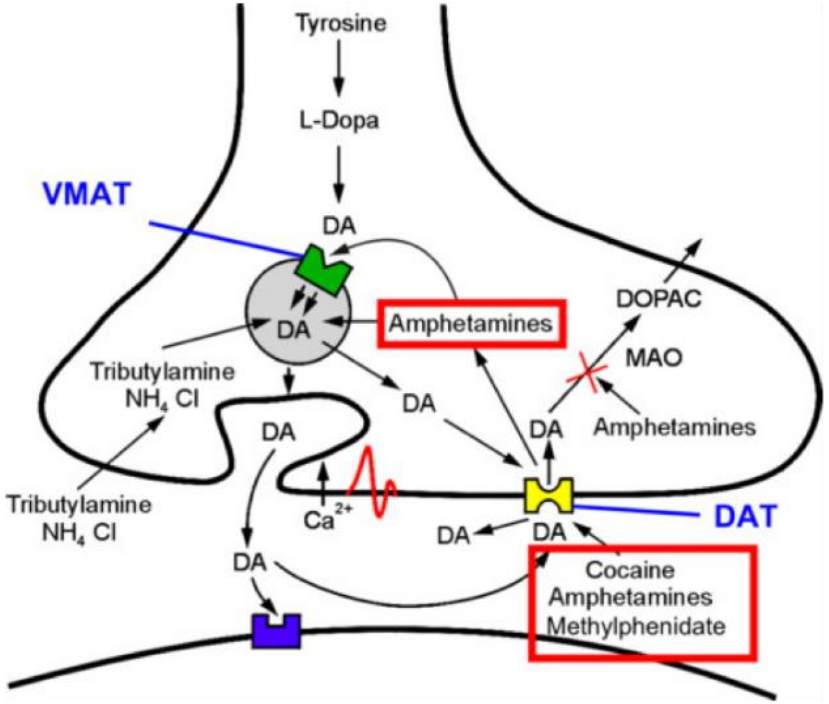


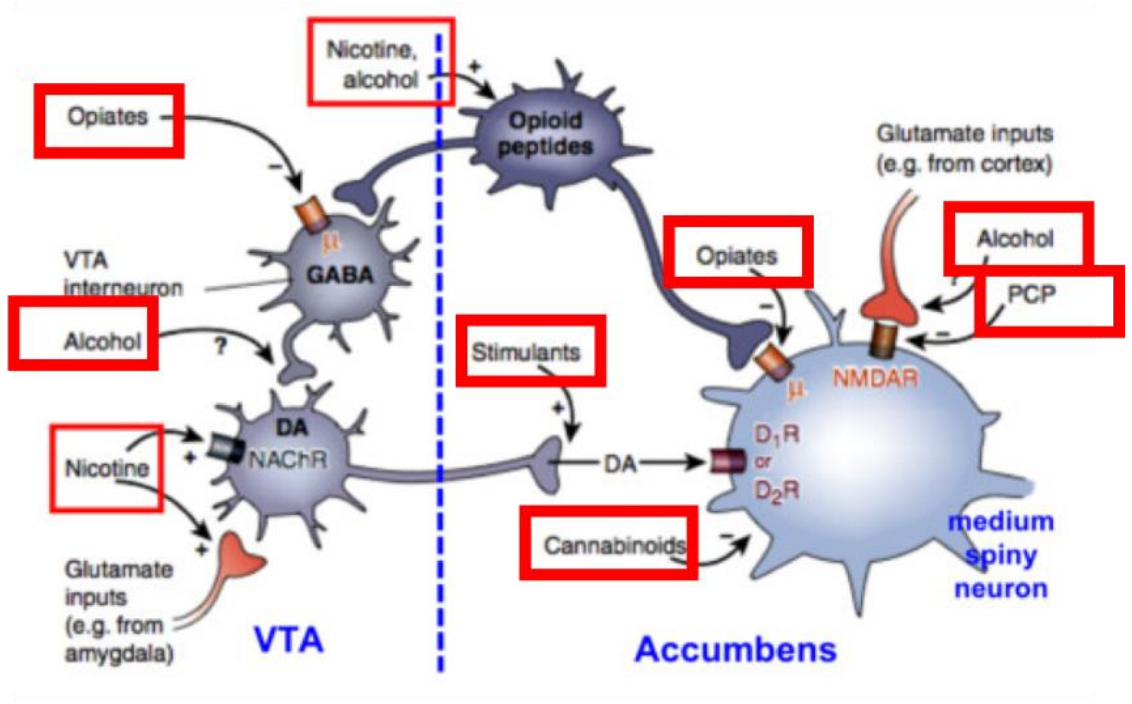
Olds and Milner, 1953



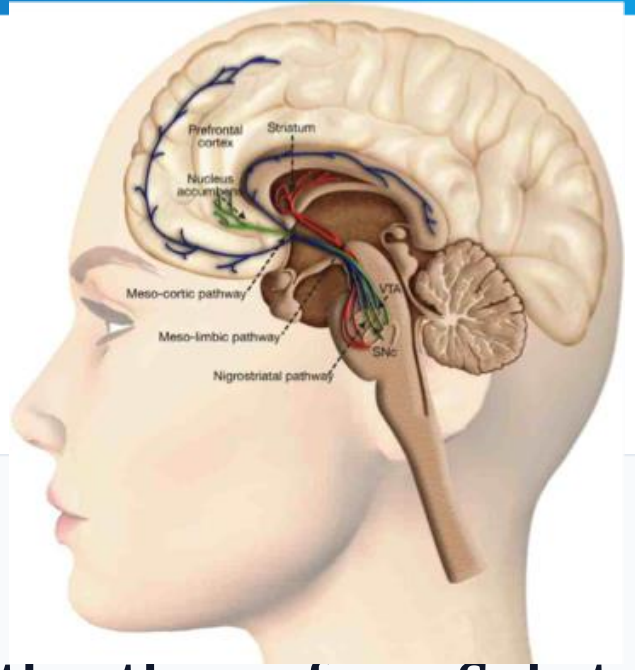

Self stimulation is effective when activating dopaminergic pathways
(Midbrain Forebrain Bundle, MFB)
a.k.a reward circuits

# Striatal dopamine: bottleneck for addiction



# Midbrain Dopamine pathways




## Mesolimbic pathway from Ventral Tegmental Area (VTA)

Projects primarily to the ventral striatum (nucleus accumbens) and prefrontal cortex. Heavily involved in reward processing, motivation, and assigning value to environmental cues.

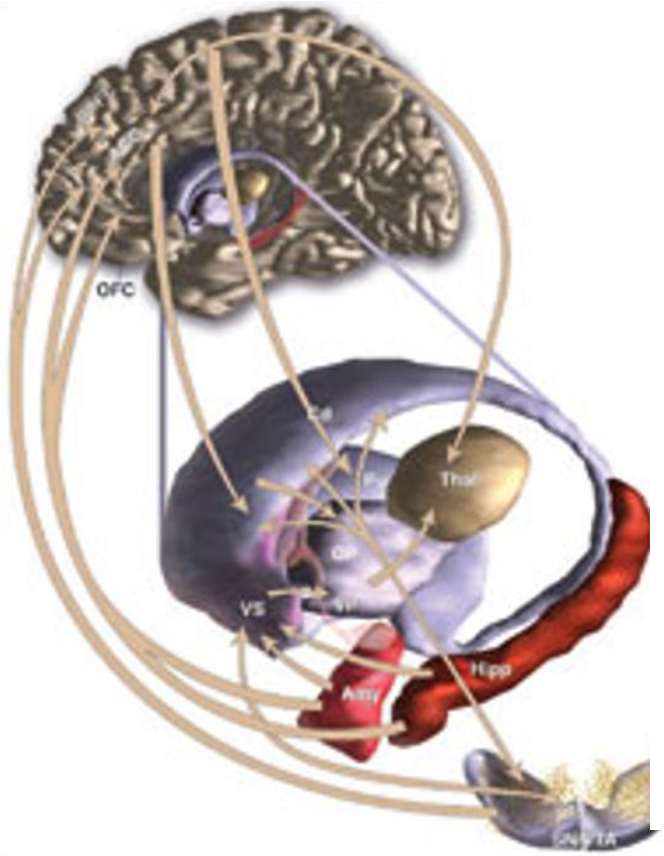
## Nigro-striatl pathway from Substantia Nigra pars compacta (SNc)

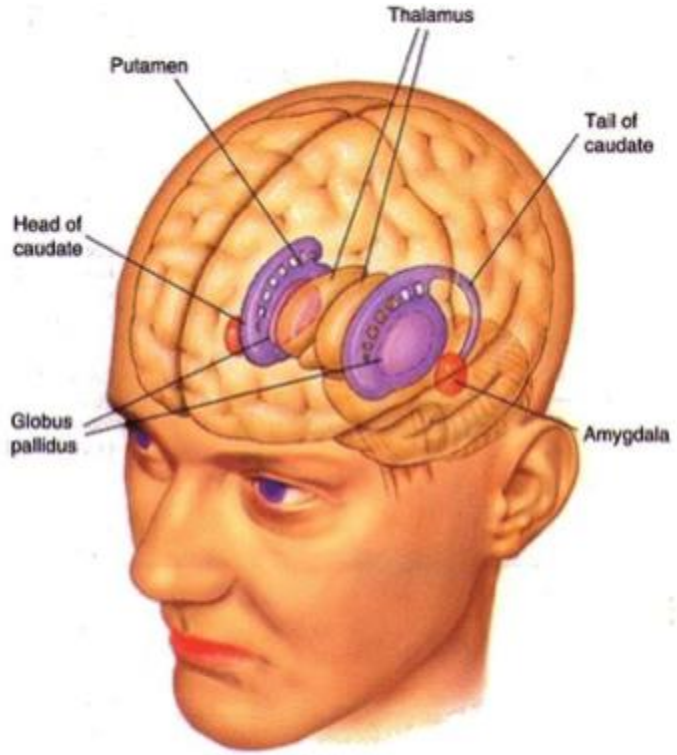
Projects primarily to the dorsal striatum. Crucial for motor control, habit formation, and the initiation of goal-directed movements.

# Basal Ganglia anatomy supporting RL

## Cortico-Basal Ganglia-Thalamo-Cortical loop

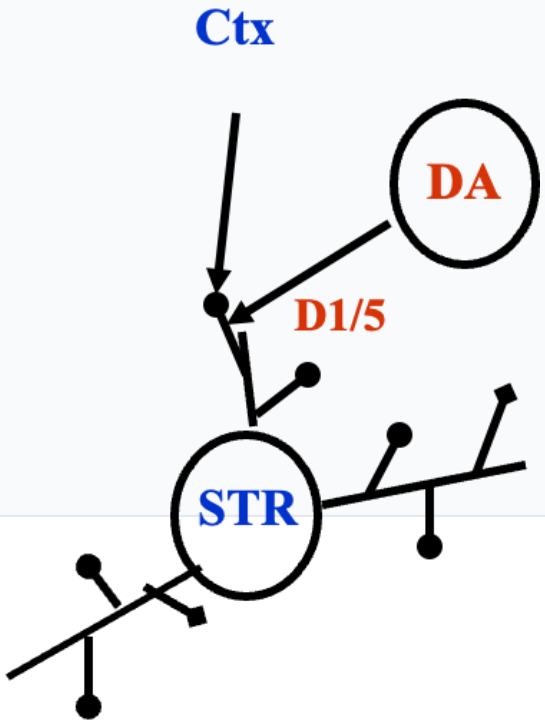
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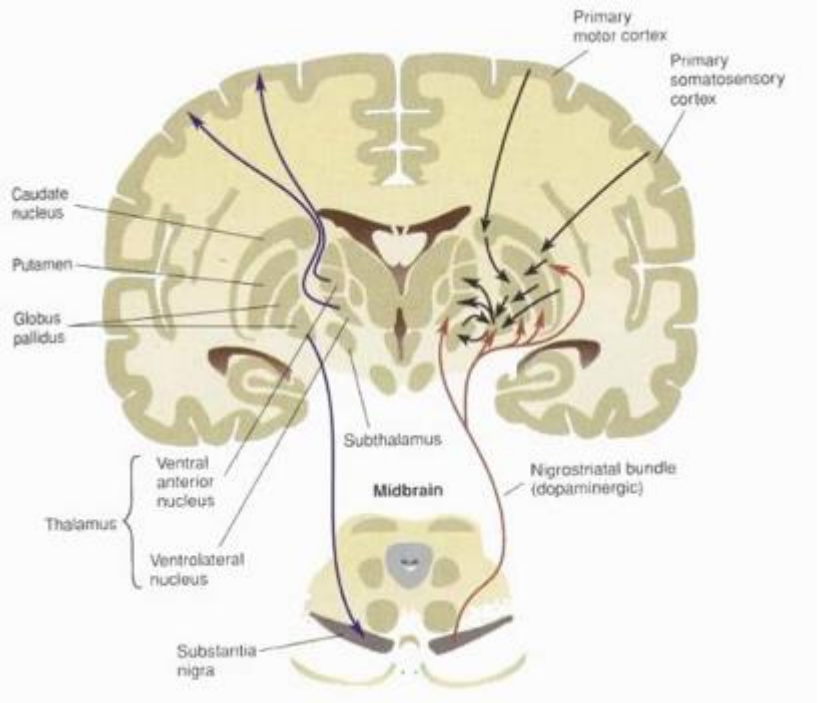


# Dopamine within Basal Ganglia anatomy

**Terminating on cortico-striatal synapses**



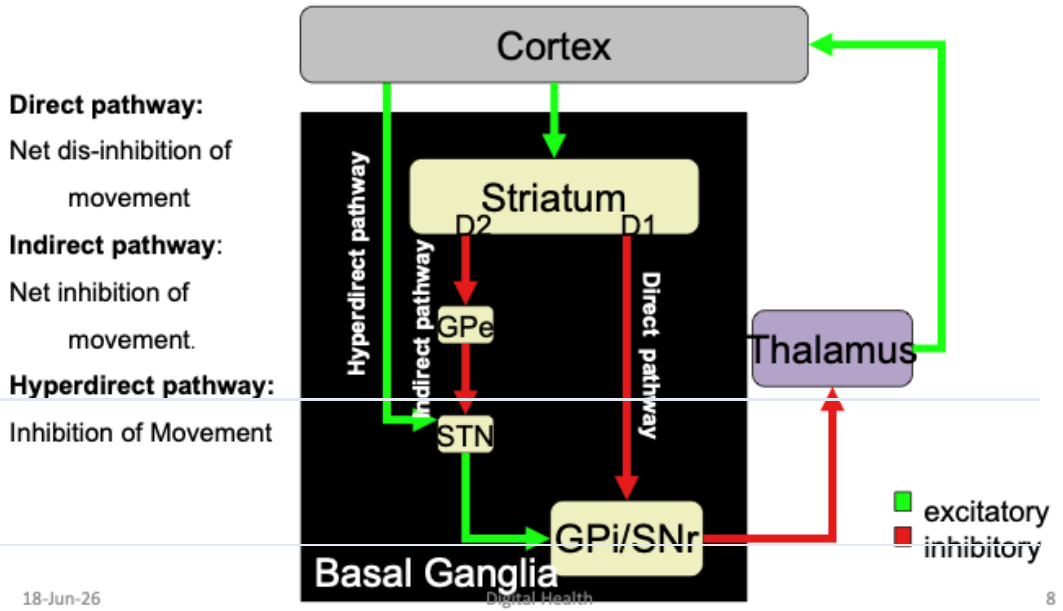



# Circuitry & Plasticity

## The Action Gatekeeper

The basal ganglia continuously orchestrate movement through a delicate balance of the direct (D1) and indirect (D2) receptor pathways.

Phasic dopamine acts as the gatekeeper, driving plasticity to selectively reinforce rewarded actions.



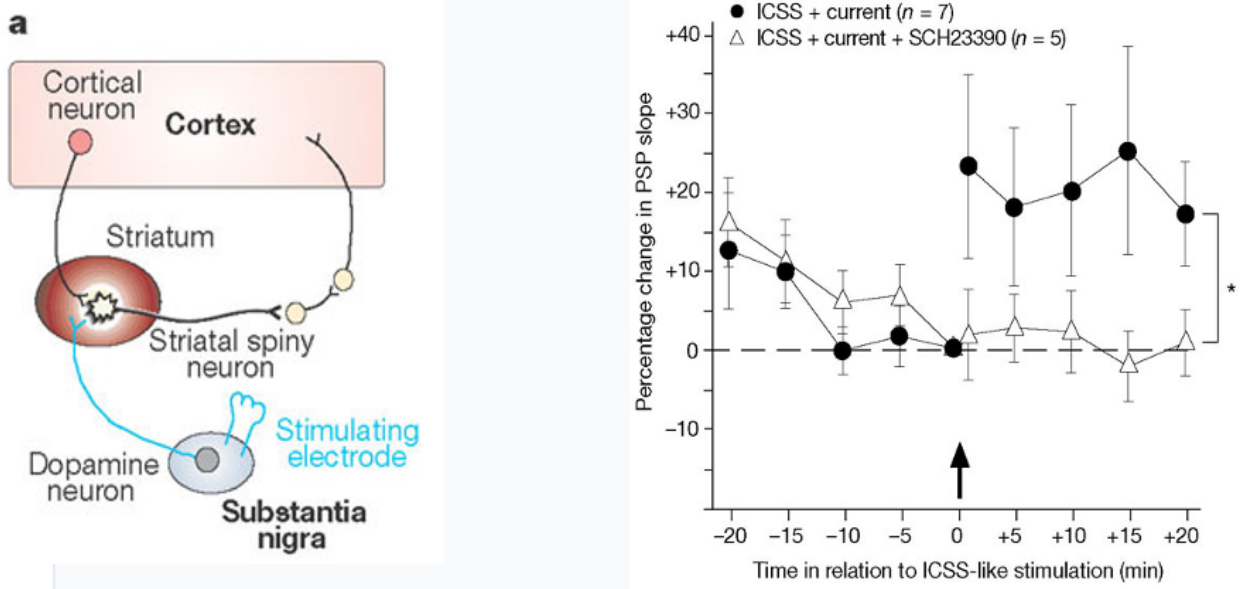

| Pathway | Receptor Affinity | Effect on Thalamus | Behavioral Outcome |
|---|---|---|---|
| Direct | D1 (Excitatory to DA) | Disinhibition (Excites) | Facilitates Action (Go) |
| Indirect | D2 (Inhibitory to DA) | Inhibition (Suppresses) | Suppresses Action (No-Go) |

# Cortico-Striatal Plasticity

## The Three-Factor Rule

Learning requires the simultaneous convergence of three signals at the striatal spiny projection neuron (SPN).

We need cortical glutamate (representing the state/action), SPN depolarization, and crucially, a phasic burst of dopamine to induce Long-Term Potentiation in (LTP) in indirect pathway.




*Reynolds eg al., Nature 2001*

# Dopamine TD error hypothesis




## Schultz, Dayan, & Montague (1997)

VTA, SNc  recordings demonstrated that midbrain dopamine neurons quantitatively encode the TD error in vivo.



They burst to unexpected rewards, shift their firing to the earliest predictive cue, and critically, pause when an expected reward is omitted.

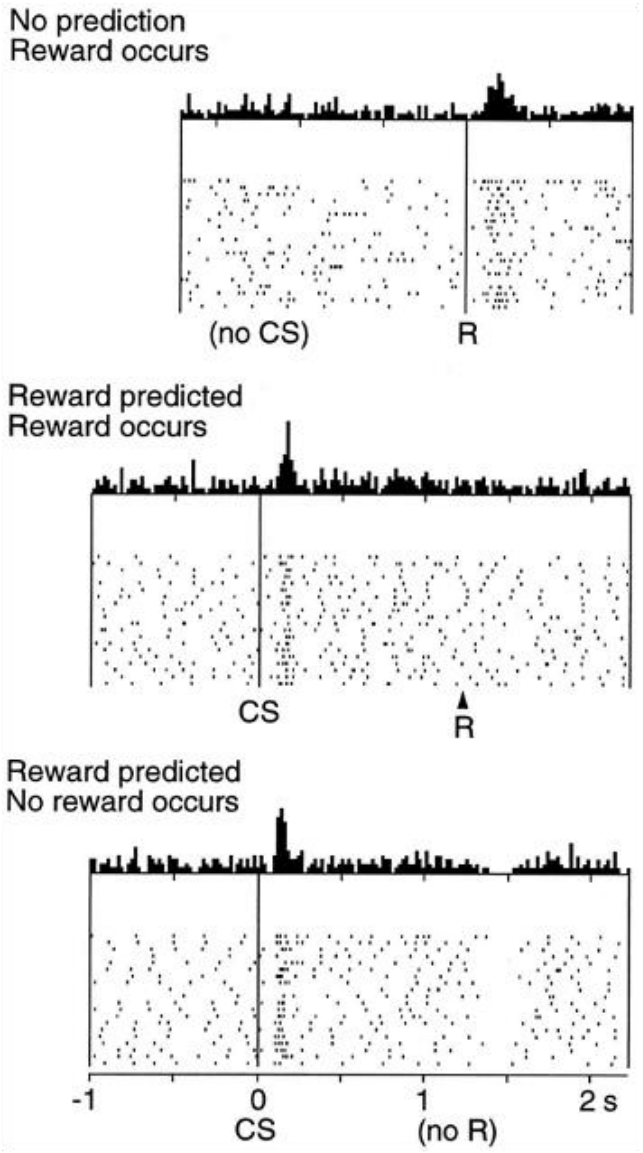


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*Schultz et al, JNS 13: 900-913 ,1993*

# The Canonical Paradigm: A Summary

- **Algorithm:** Temporal Difference learning solves the credit assignment problem mathematically.
- **Signal:** Midbrain dopamine transients are the biological realization of the TD prediction error.
- **Target:** The basal ganglia act as the physical hardware receiving these updates.
- **Mechanism:** Dopamine-dependent plasticity at cortico-striatal synapses solidifies the learned action policies.

# Part 3: Challenges to the story

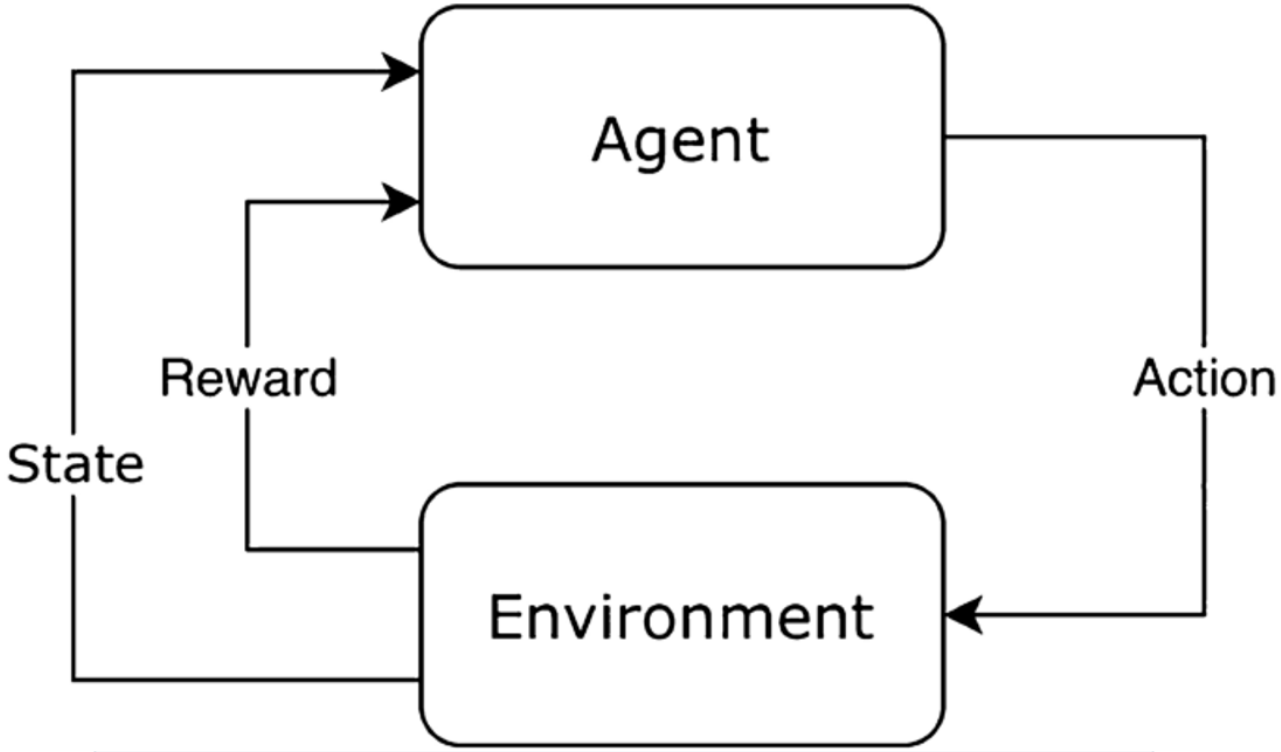
# 1. The Representation Challenge

## Defining the State

Classic models assume discrete, easily identifiable states.

Real-world environments are continuous, noisy, and high-dimensional.

How does the brain define the "state" over which prediction errors are calculated? This requires immense prefrontal and hippocampal abstraction before dopamine even fires.



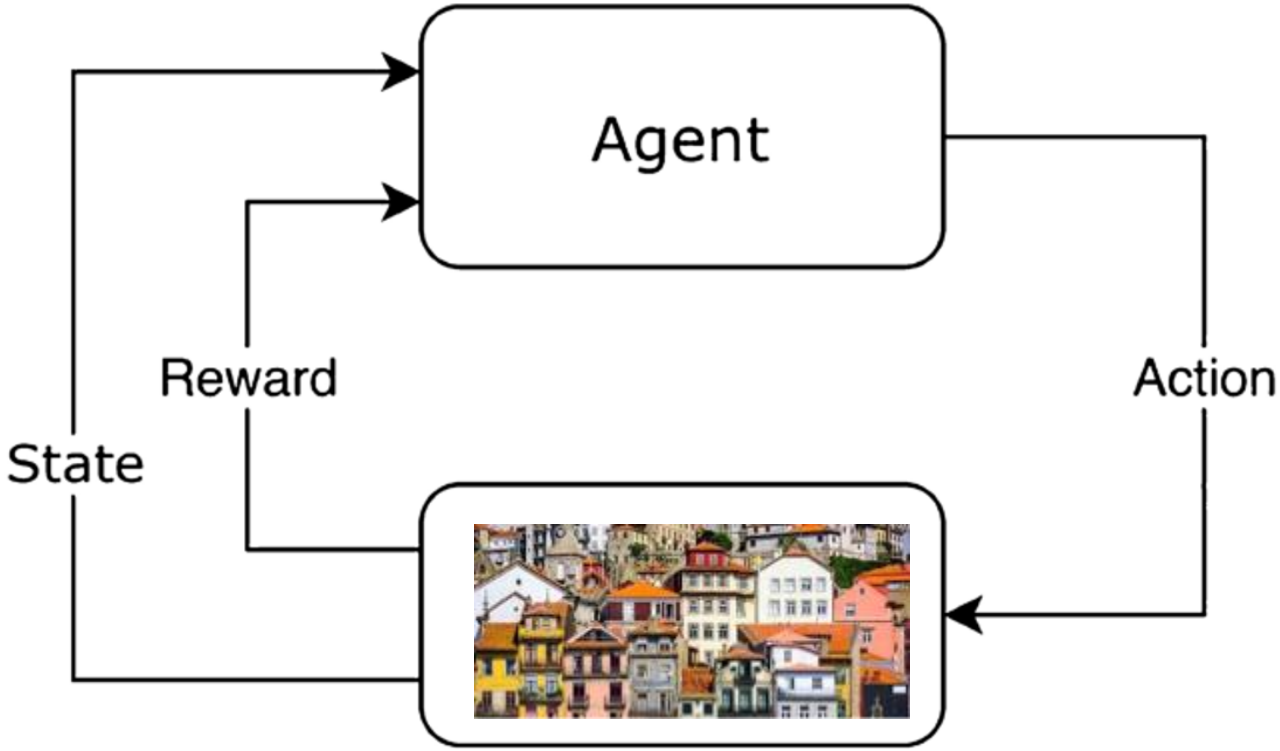
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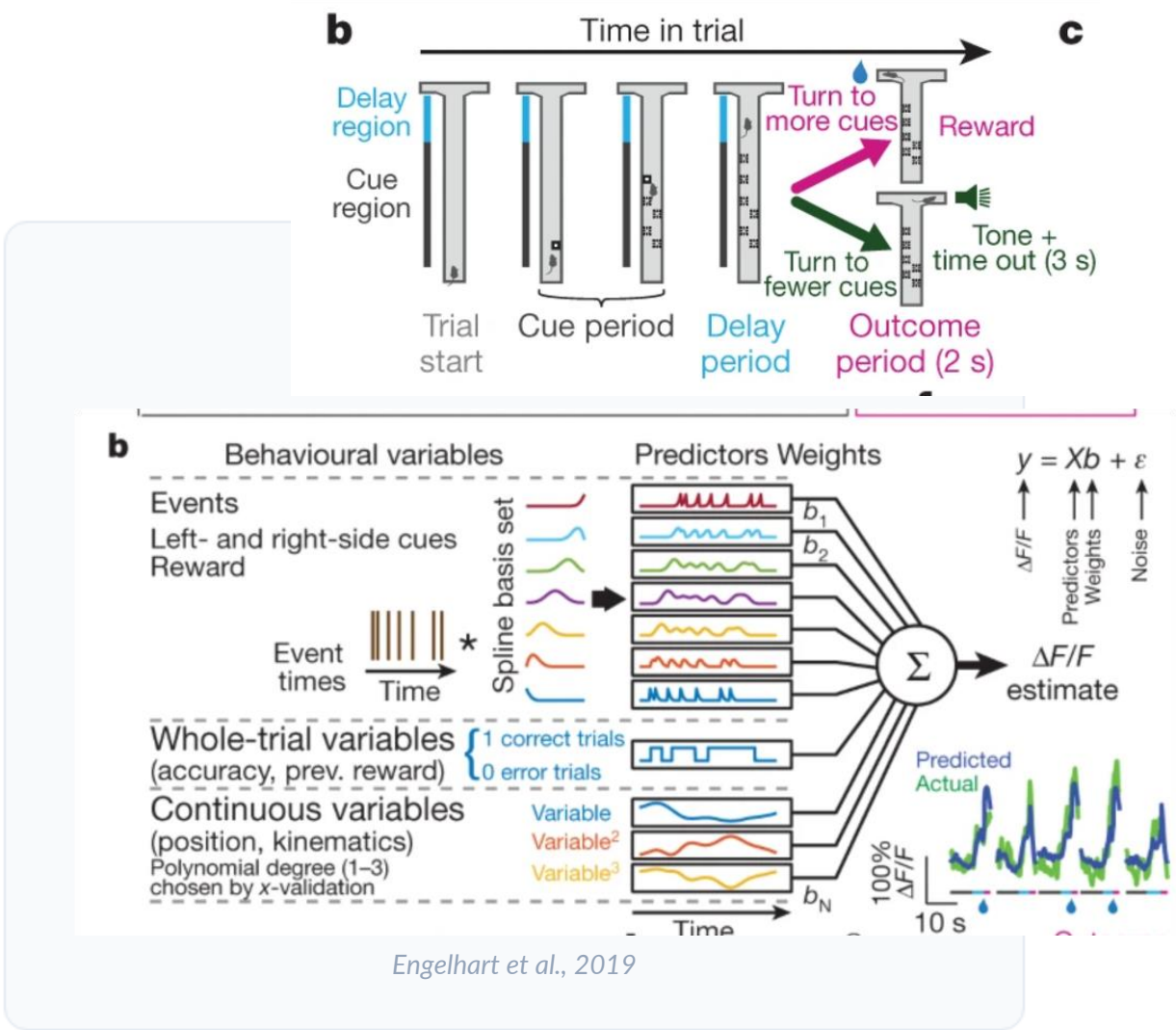

*No man ever steps in the same river twice, for it's not the same river and he's not the same man*

*Heraclitus , ~500BC*

# 2. Circuit Heterogeneity

## Not a Monolith

The dopaminergic system is highly multifunctional. Different sub-regions of the striatum receive distinctly tuned dopamine signals, which may represent different parameters




*Engelhart et al., 2019*

# Beyond Instantaneous RPE

## Slow Dynamics

Dopamine release is not exclusively composed of sub-second bursts. Slower, minute-by-minute fluctuations occur that don't easily fit a strict prediction error framework.

## State & Motivation

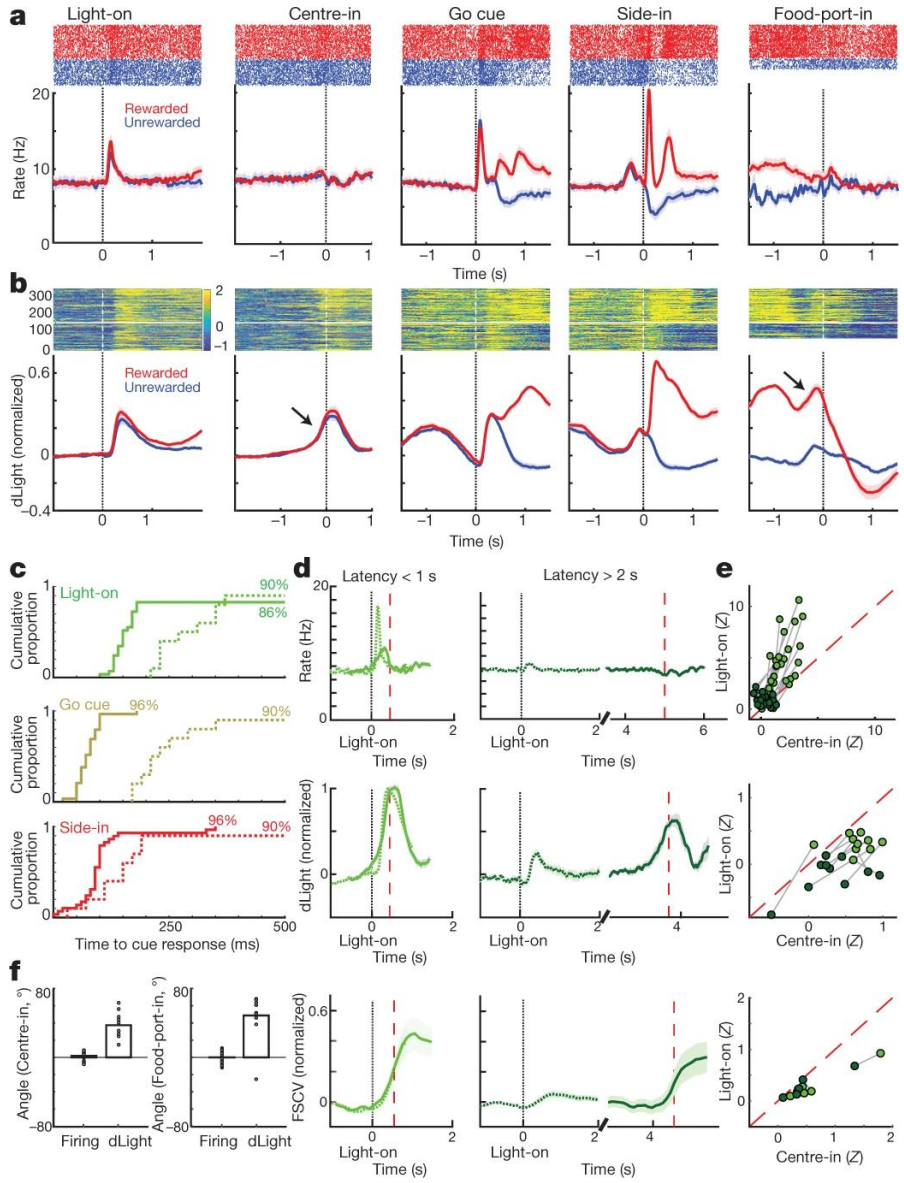
These slower dynamics often correlate with generalized arousal, motivational states, or the perceived "distance" to a distal goal during spatial navigation.

# Striatal dopamine doesn't follow VTA firing rates

## Mohebi et al., (2019/2024)

Recent literature shows dopamine function extends beyond fast prediction errors.

Strikingly, slow ramping dopamine release in the striatum correlates strongly with changing states of motivation as subjects approach a goal, independent of spiking rate.




Mohebi et al., 2019

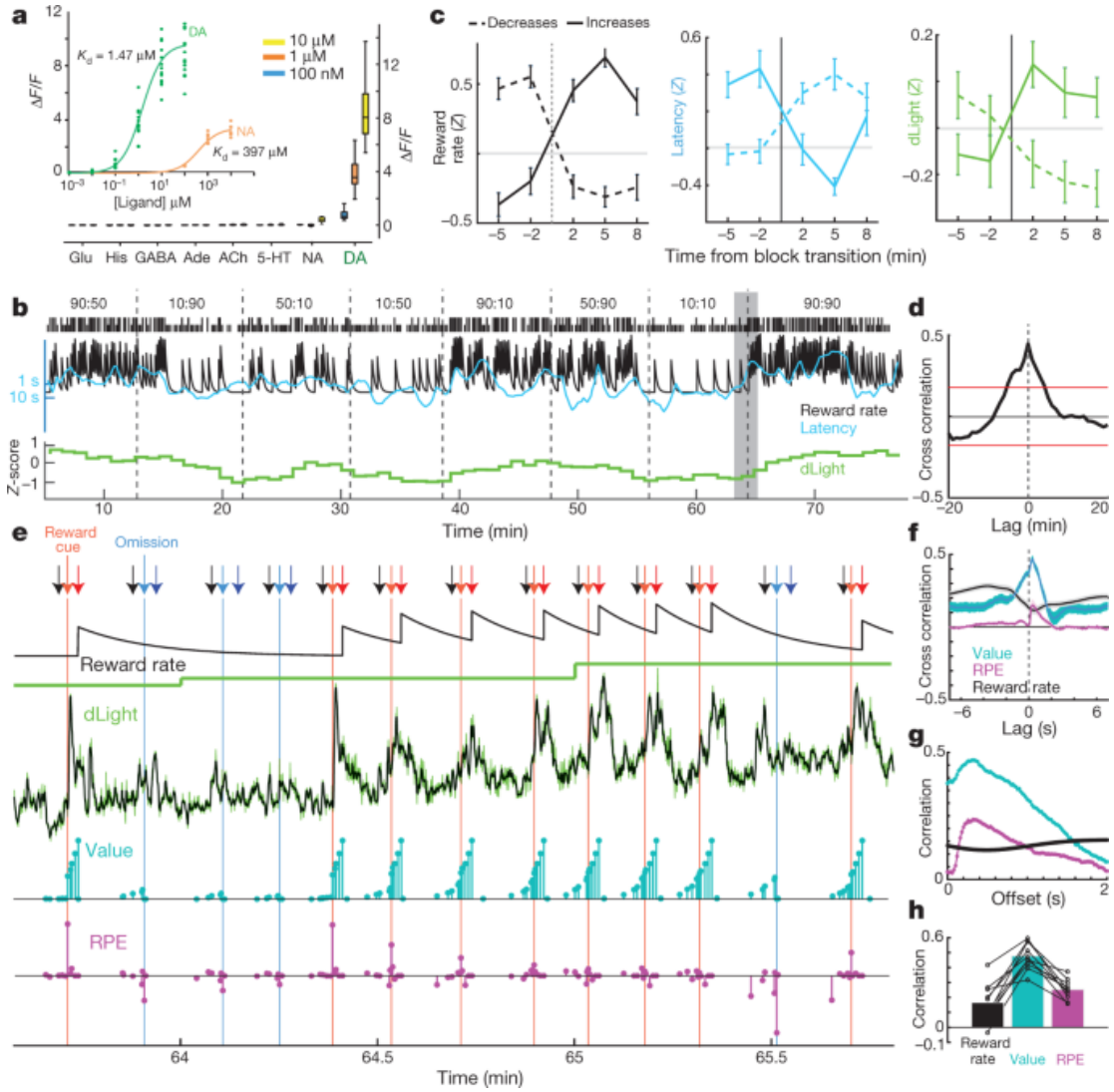
# Striatal dopamine ramps with motivation/vigor

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Different striatal subregions loot at different time horizons




Mohebi et al., 2019

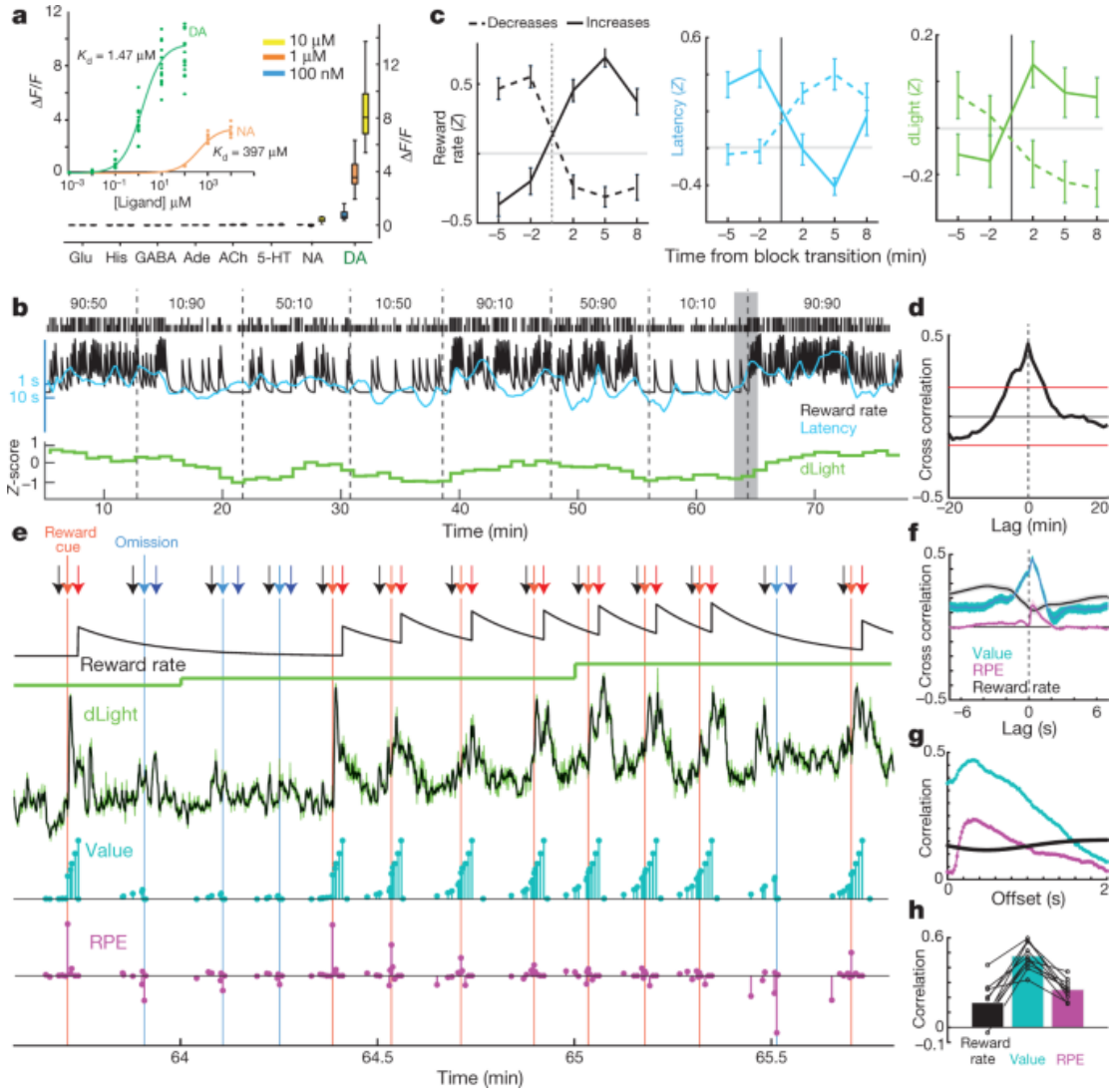
# Striatal dopamine ramps with motivation/vigor

## Mohebi et al., (2019/2024)

Recent literature shows dopamine function extends beyond fast prediction errors.

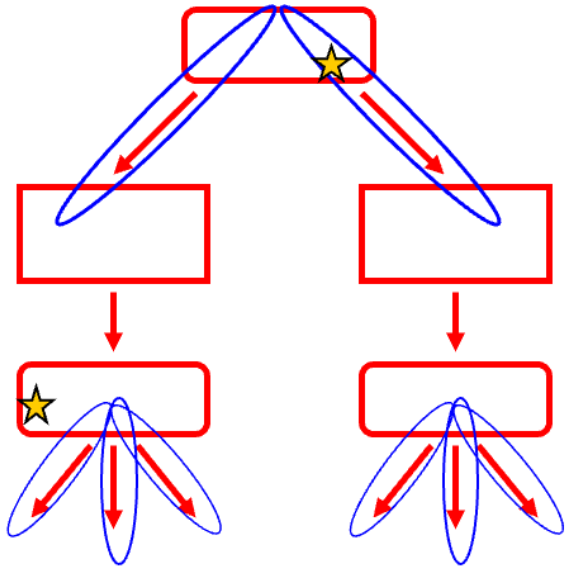
Slow ramping dopamine release in the striatum correlates strongly with changing states of motivation as subjects approach a goal, independent of spiking rate.

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Mohebi et al., 2019

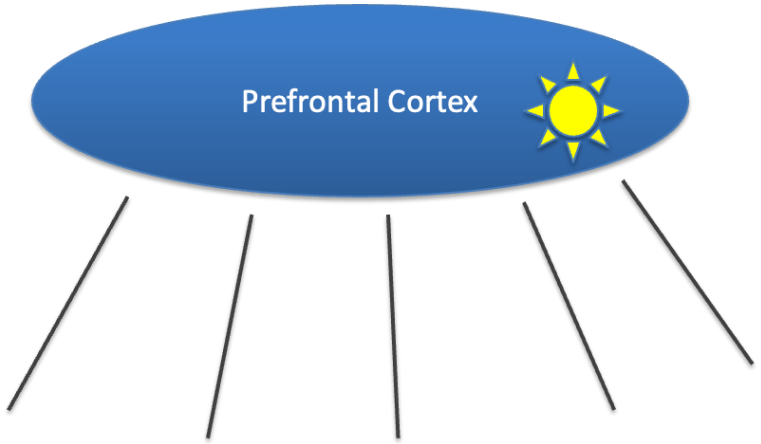
# What about model-based learning and planning?

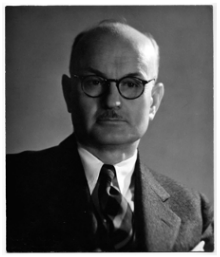


- No explicit knowledge about the future is necessary
- But what about planning?

# What about model-based learning and planning?

## Model based goal directed behavior



*Edward Tolman (1948)*
*Cognitive maps of rats and men*

# What about model-based learning and planning?



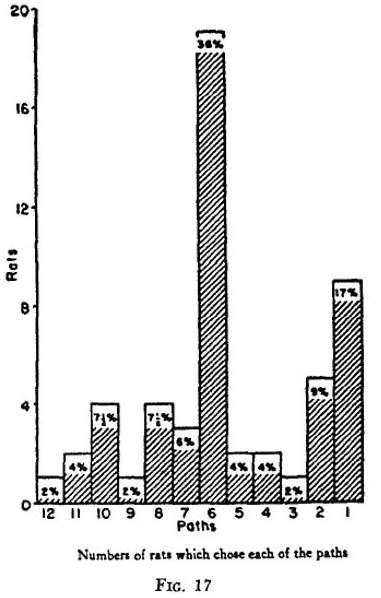
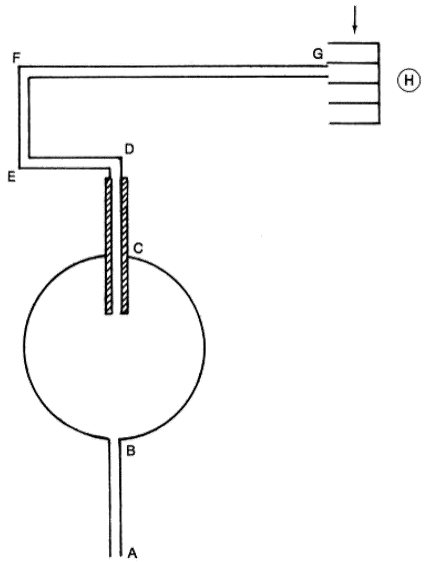


FIG. 17




- Not everything is stimulus response
- Rather, there are internal representations
- **Cognitive maps** as an example of an internal/mental representation

Tolman EC (1948) Cognitive maps in rats and men. Psychol Rev 55:189-208.

# What about model-based learning and planning?



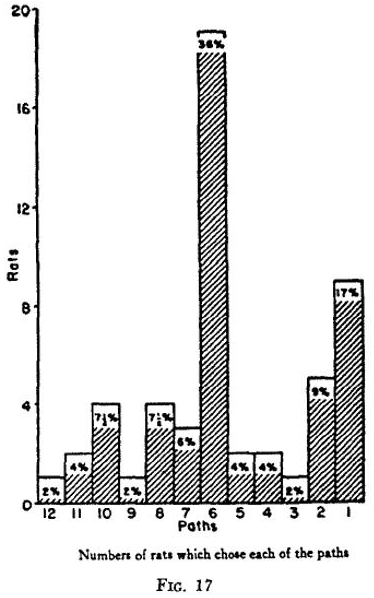
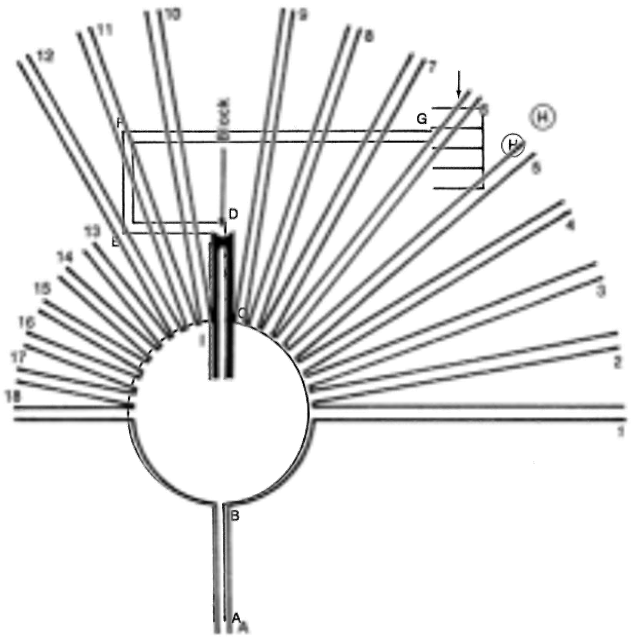


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